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Machine Learning: III. Principles of Regularization

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robot learning



Outline

1. Overfitting and Underfitting
2. Bias-Variance Tradeoff
3. Regularization
4. Conclusion



Outline

1 Overfitting and Underfitting

2 Bias-Variance Tradeoff

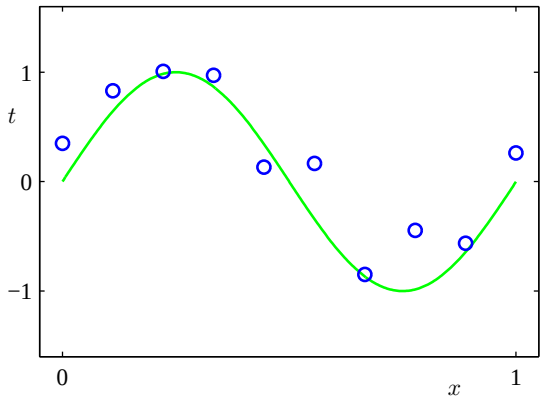
3 Regularization

4 Conclusion



Example: Nonlinear regression

Consider a curve fitting example where we are given a labeled data set of N examples $\langle (x_i, y_i) \rangle_{i=1}^{N}$. Labels are generated from the **target function** $\sin(2\pi x)$ plus a bit of Gaussian noise.



Polynomial Regression

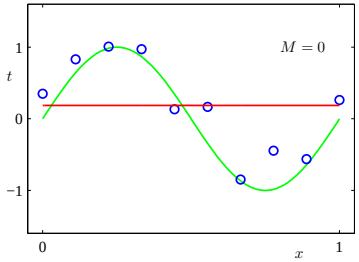
For $x \in \mathbb{R}^1$ the polynomial prediction model of degree M is:

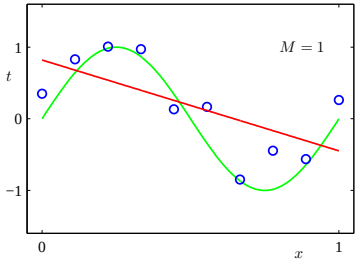
$$\hat{y} = f(x, \theta) = \theta_0 + \theta_1 x^1 + \theta_2 x^2 + ... + \theta_M x^M = \sum_{j=0}^{M} \theta_j x^j$$

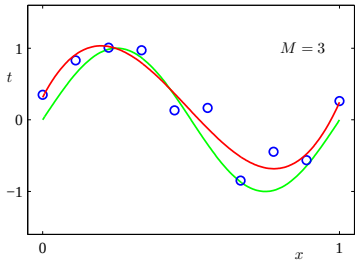
The optimal $\overset{\star}{\theta}$ are learned by minimizing the **empirical risk**:

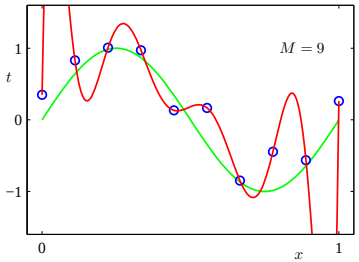
$$\overset{\star}{\theta} := \arg\min_{\theta} \sum_{i=1}^{N} (f(x_i, \theta) - y_i)^2$$

How to choose M?



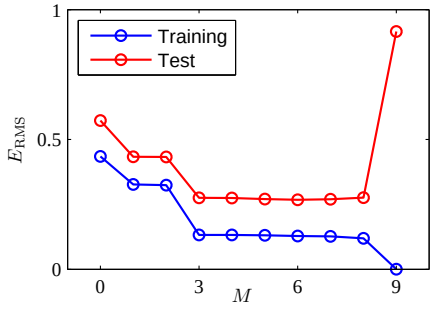






Generalization Performance

- **Overfitting**: Model perfectly fits the training data (incl. noise)
- **Underfitting**: Model fails to fit the training data
- **Generalization**: Model is accurate on test data



Questions to Answer for Yourself / Discuss with Friends

Ponder: Describe how you could tell if a model is overfitting, underfitting, or generalizing well using only training and test data results. What metrics or observations would you look at?

Repetition: How is the model complexity related to under-, overfitting?

Questions to Answer for Yourself / Discuss with Friends

Ponder: Describe how you could tell if a model is overfitting, underfitting, or generalizing well using only training and test data results. What metrics or observations would you look at?

- **Underfitting:** Both training and test errors descend, but get stuck.
- **Generalizing:** Both training and test errors descend and approach a minimum.
- **Overfitting:** Training error keeps getting smaller, while test error starts to increase.

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Repetition: How is the model complexity related to under-, overfitting?

More complex models are able to ”memorize” the training data, including the noise. Restricting a model complexity is usually needed for it to generalize well to unseen data. Restricting it too much, however, could lead to underfitting.

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1 Overfitting and Underfitting

2 Bias-Variance Tradeoff

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Preliminary - Expected Value

z is a **continuous** r.v. $\quad\rightsquigarrow\quad \mathrm{E}[z] = \int_z z \ \mathrm{p}(z) \ \mathrm{d}z$

z is a c.r.v. **conditional** to v $\quad\rightsquigarrow\quad \mathop{\mathrm{E}}_{z|v}[z] = \int_z z \ \mathrm{p}(z \mid v) \ \mathrm{d}z$

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x	y
m	180
m	170
f	160
f	170
m	170
f	160
m	170

Table 1: Gender (x) and Height (y)

z is a **discrete** r.v. $\rightsquigarrow$

$$\mathrm{E}[z] = \sum_z z \ \mathrm{p}(z) \qquad \mathop{\mathrm{E}}_{z|v}[z] = \sum_z z \ \mathrm{p}(z \mid v)$$

$$\mathrm{E}[y] = 160\frac{2}{7} + 170\frac{4}{7} + 180\frac{1}{7} = 168.6$$

$$\mathop{\mathrm{E}}_{y|x=f}[y] = 160\frac{2}{3} + 170\frac{1}{3} + 180\frac{0}{3} = 163.3$$

Preliminary - Properties of Expectation

- Constant $\alpha \in \mathbb{R}$:

$$\mathrm{E}[\alpha\ z] = \int_z \alpha\ z\ \mathrm{p}(z)\ \mathrm{d}z = \alpha \int_z z\, p(z)\, dz = \alpha\ \mathrm{E}[z]$$

- Linearity of expectation:

$$\mathrm{E}[\alpha z + \beta v] = \alpha\, \mathrm{E}[z] + \beta\, \mathrm{E}[v]$$

- Expectation of two uncorrelated r.v.:

$$\mathop{\mathrm{E}}_{z,v}[z\ v] = \mathop{\mathrm{E}}_{z}[\mathop{\mathrm{E}}_{v}[z\ v]] = \mathop{\mathrm{E}}_{z}[z\ \mathop{\mathrm{E}}_{v}[v]] = \mathop{\mathrm{E}}_{v}[v]\ \mathop{\mathrm{E}}_{z}[z]$$

- Expectation of two correlated r.v.:

$$\mathop{\mathrm{E}}_{z,v}[z\ v] = \mathop{\mathrm{E}}_{z}[\mathop{\mathrm{E}}_{v|z}[z\ v]] = \mathop{\mathrm{E}}_{z}[z\ \mathop{\mathrm{E}}_{v|z}[v]]$$

Expected Target

- A training dataset $\mathcal{D} \coloneqq \langle (x_i, y_i) \rangle_{i=1}^{N}$ drawn i.i.d. from a distribution $\mathcal{P}$

- **Expected target** of y given x

$$\bar{y}(x) = \mathop{\mathrm{E}}_{y|x}[y] = \int_y y \ \mathrm{p}(y \mid x) \ \mathrm{d}y$$

Estimated Prediction Model (Regression)

- ML **estimates a prediction model** $\hat{f}(x,\, \theta)$ by minimizing:

$$\mathop{\mathrm{E}}_{(x,\, y) \sim \mathcal{P}} \left[\left(\hat{f}(x,\, \theta) - y \right)^2 \right] = \int_x \int_y \left(\hat{f}(x,\, \theta) - y \right)^2 \, \mathrm{p}(x,\, y) \, \mathrm{d}y \, \mathrm{d}x$$

- As $\mathrm{p}(x,\, y)$ is typically unknown we **approximate the error**:

$$\mathop{\mathrm{E}}_{(x,\, y) \sim \mathcal{P}} \left[\left(\hat{f}(x,\, \theta) - y \right)^2 \right] \approx \frac{1}{N} \sum_{i=1}^{N} \left(\hat{f}(x_i,\, \theta) - y_i \right)^2$$

- The **estimated model** $\hat{f}$ given a training set $\mathcal{D} \coloneqq \langle (x_i,\, y_i) \rangle_{i=1}^{N}$

$$\hat{f}(x;\, \mathcal{D}) = \hat{f}(x,\, \overset{\star}{\theta}) \quad \text{s.t.} \quad \overset{\star}{\theta} = \arg\min_{\theta} \frac{1}{N} \sum_{i=1}^{N} \left(\hat{f}(x_i,\, \theta) - y_i \right)^2$$

Expected Test Error and Prediction Model

- **Expected test error** of $\hat{f}(x;\, D)$ **given a training set** $\mathcal{D}$:

$$\mathop{\mathrm{E}}_{(x,\,y)\sim\mathcal{P}}\left[\left(\hat{f}(x;\,\mathcal{D})-y\right)^2\right]=\int_x\int_y\left(\hat{f}(x;\,\mathcal{D})-y\right)^2\mathrm{p}(x,\,y)\,\mathrm{d}y\,\mathrm{d}x$$

- Since $(x,\,y)\sim\mathcal{P}$ then $\mathcal{D}:=\langle(x_i,\,y_i)\rangle_{i=1}^N\sim\mathcal{P}^N$ is also a r.v.

- The **expected prediction model** over sampled datasets is:

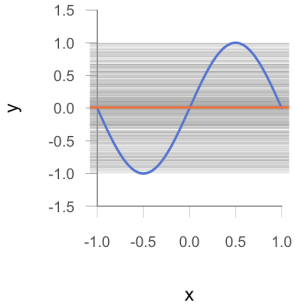
$$\bar{f}(x):=\mathop{\mathrm{E}}_{\mathcal{D}\sim\mathcal{P}^N}\left[\hat{f}(x;\,\mathcal{D})\right]=\int_{\mathcal{D}}\hat{f}(x;\,\mathcal{D})\;\mathrm{p}(\mathcal{D})\,\mathrm{d}\mathcal{D}$$

Expected Prediction Model

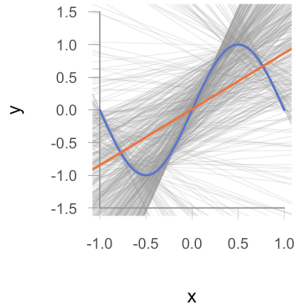
- The **expected prediction model** over sampled datasets is:

$$\bar{f}(x) \coloneqq \mathop{\mathrm{E}}_{\mathcal{D}\sim\mathcal{P}^N}\left[\hat{f}(x;\,\mathcal{D})\right] = \int_{\mathcal{D}} \hat{f}(x;\,\mathcal{D})\;\mathrm{p}(\mathcal{D})\,\mathrm{d}\mathcal{D}$$

$f(x) = \theta_0$



$f(x) = \theta_0 + \theta_1 x$



Source: `allmodelsarewrong.github.io`

Expected Test Error

- **Expected test error** of $\hat{f}(x;\, \mathcal{D})$ **over training sets** $\mathcal{D}$:

$$\mathop{\mathrm{E}}_{\substack{(x,\,y)\sim\mathcal{P} \\ \mathcal{D}\sim\mathcal{P}^N}} \left[\left(\hat{f}(x;\, \mathcal{D}) - y \right)^2 \right] = \int_x \int_y \int_{\mathcal{D}} \left(\hat{f}(x;\, \mathcal{D}) - y \right)^2 \mathrm{p}(x,\, y)\ \mathrm{p}(\mathcal{D})\ \mathrm{d}\mathcal{D}\, \mathrm{d}y\, \mathrm{d}x$$

- The bias-variance decomposition shows how this expected test error is expressed as a sum of:
  - **Bias**: How well does the prediction model fits the target?
  - **Variance**: How much do the predictions of models vary when trained on different sampled training sets?
  - **Noise**: How much of the target variable cannot be unexplained?

Bias-Variance Tradeoff (I)

$$
\begin{aligned}
\mathop{\mathrm{E}}_{x,y,\mathcal{D}}\left[\left(\hat{f}(x;\,\mathcal{D})-y\right)^2\right] &= \mathop{\mathrm{E}}_{x,y,\mathcal{D}}\left[\left(\hat{f}(x;\,\mathcal{D})-\bar{f}(x)+\bar{f}(x)-y\right)^2\right] \\
&= \mathop{\mathrm{E}}_{x,y,\mathcal{D}}\left[\left(\left(\hat{f}(x;\,\mathcal{D})-\bar{f}(x)\right)+\left(\bar{f}(x)-y\right)\right)^2\right] \\
&= \mathop{\mathrm{E}}_{x,y,\mathcal{D}}\Bigg[\left(\hat{f}(x;\,\mathcal{D})-\bar{f}(x)\right)^2 \\
&\qquad\qquad +\left(\bar{f}(x)-y\right)^2 \\
&\qquad\qquad +2\left(\hat{f}(x;\,\mathcal{D})-\bar{f}(x)\right)\left(\bar{f}(x)-y\right)\Bigg]
\end{aligned}
$$

- Leads to:

$$
\begin{aligned}
\mathop{\mathrm{E}}_{x,y,\mathcal{D}}\left[\left(\hat{f}(x;\,D)-y\right)^2\right] &= \mathop{\mathrm{E}}_{x,\mathcal{D}}\left[\left(\hat{f}(x;\,\mathcal{D})-\bar{f}(x)\right)^2\right]+\mathop{\mathrm{E}}_{x,y}\left[\left(\bar{f}(x)-y\right)^2\right] \\
&\quad +2\mathop{\mathrm{E}}_{x,y,\mathcal{D}}\left[\left(\hat{f}(x;\,\mathcal{D})-\bar{f}(x)\right)\left(\bar{f}(x)-y\right)\right]
\end{aligned}
$$

Bias-Variance Tradeoff (II)

$$\mathop{\mathrm{E}}_{x,y,\mathcal{D}}\left[\left(\hat{f}(x;\,\mathcal{D})-y\right)^2\right] = \mathop{\mathrm{E}}_{x,\mathcal{D}}\left[\left(\hat{f}(x;\,\mathcal{D})-\bar{f}(x)\right)^2\right] + \mathop{\mathrm{E}}_{x,y}\left[\left(\bar{f}(x)-y\right)^2\right] + \underbrace{2\mathop{\mathrm{E}}_{x,y,\mathcal{D}}\left[\left(\hat{f}(x;\,\mathcal{D})-\bar{f}(x)\right)\left(\bar{f}(x)-y\right)\right]}_{\text{Cancels out}}$$

$$\begin{aligned}
&\mathop{\mathrm{E}}_{x,y}\left[\mathop{\mathrm{E}}_{\mathcal{D}}\left[\left(\hat{f}(x;\,\mathcal{D})-\bar{f}(x)\right)\left(\bar{f}(x)-y\right)\right]\right] \\
&= \mathop{\mathrm{E}}_{x,y}\left[\left(\bar{f}(x)-y\right)\left(\mathop{\mathrm{E}}_{\mathcal{D}}\left[\hat{f}(x;\,\mathcal{D})-\bar{f}(x)\right]\right)\right] \\
&= \mathop{\mathrm{E}}_{x,y}\left[\left(\bar{f}(x)-y\right)\left(\bar{f}(x)-\bar{f}(x)\right)\right] = 0
\end{aligned}$$

Bias-Variance Tradeoff (II)

$$\mathop{\mathrm{E}}_{x,y,\mathcal{D}}\left[\left(\hat{f}(x;\,\mathcal{D})-y\right)^2\right] = \mathop{\mathrm{E}}_{x,\mathcal{D}}\left[\left(\hat{f}(x;\,\mathcal{D})-\bar{f}(x)\right)^2\right] + \mathop{\mathrm{E}}_{x,y}\left[\left(\bar{f}(x)-y\right)^2\right] + \underbrace{2\mathop{\mathrm{E}}_{x,y,\mathcal{D}}\left[\left(\hat{f}(x;\,\mathcal{D})-\bar{f}(x)\right)\left(\bar{f}(x)-y\right)\right]}_{\text{Cancels out}}$$

$$\begin{aligned}
&\mathop{\mathrm{E}}_{x,y}\left[\mathop{\mathrm{E}}_{\mathcal{D}}\left[\left(\hat{f}(x;\,\mathcal{D})-\bar{f}(x)\right)\left(\bar{f}(x)-y\right)\right]\right] \\
&= \mathop{\mathrm{E}}_{x,y}\left[\left(\bar{f}(x)-y\right)\left(\mathop{\mathrm{E}}_{\mathcal{D}}\left[\hat{f}(x;\,\mathcal{D})-\bar{f}(x)\right]\right)\right] \\
&= \mathop{\mathrm{E}}_{x,y}\left[\left(\bar{f}(x)-y\right)\left(\bar{f}(x)-\bar{f}(x)\right)\right] = 0
\end{aligned}$$

- So far we decomposed the estimated test error to:

$$\mathop{\mathrm{E}}_{x,y,\mathcal{D}}\left[\left(\hat{f}(x;\,\mathcal{D})-y\right)^2\right] = \underbrace{\mathop{\mathrm{E}}_{x,\mathcal{D}}\left[\left(\hat{f}(x;\,\mathcal{D})-\bar{f}(x)\right)^2\right]}_{\text{Variance}} + \underbrace{\mathop{\mathrm{E}}_{x,y}\left[\left(\bar{f}(x)-y\right)^2\right]}_{\text{Reduce further …}}$$

Bias-Variance Tradeoff (III)

$$
\begin{aligned}
\mathop{\mathrm{E}}_{x,y}\left[\left(\bar{f}(x)-y\right)^2\right] &= \mathop{\mathrm{E}}_{x,y}\left[\left(\bar{f}(x)-\bar{y}(x)+\bar{y}(x)-y\right)^2\right] \\
&= \mathop{\mathrm{E}}_{x,y}\left[\left(\bar{f}(x)-\bar{y}(x)\right)^2\right] \\
&\quad + \mathop{\mathrm{E}}_{x,y}\left[\left(\bar{y}(x)-y\right)^2\right] \\
&\quad + \underbrace{\mathop{\mathrm{E}}_{x,y}\left[2\left(\bar{f}(x)-\bar{y}(x)\right)\left(\bar{y}(x)-y\right)\right]}_{\text{cancels out}}
\end{aligned}
$$

$$
\begin{aligned}
\mathop{\mathrm{E}}_{x,y}\left[\left(\bar{f}(x)-\bar{y}(x)\right)\left(\bar{y}(x)-y\right)\right] &= \mathop{\mathrm{E}}_{x}\left[\mathop{\mathrm{E}}_{y|x}\left[\left(\bar{f}(x)-\bar{y}(x)\right)\left(\bar{y}(x)-y\right)\right]\right] \\
&= \mathop{\mathrm{E}}_{x}\left[\left(\bar{f}(x)-\bar{y}(x)\right)\mathop{\mathrm{E}}_{y|x}\left[\left(\bar{y}(x)-y\right)\right]\right]
\end{aligned}
$$

This term cancels out:

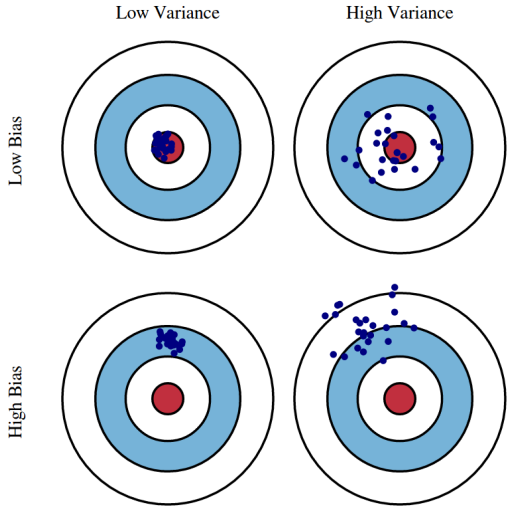
$$
\mathop{\mathrm{E}}_{y|x}\left[\bar{y}(x)-y\right] = \mathop{\mathrm{E}}_{y|x}\left[\bar{y}(x)\right] - \mathop{\mathrm{E}}_{y|x}\left[y\right] = \bar{y}(x)-\bar{y}(x) = 0
$$

Bias-Variance Tradeoff (Finale)

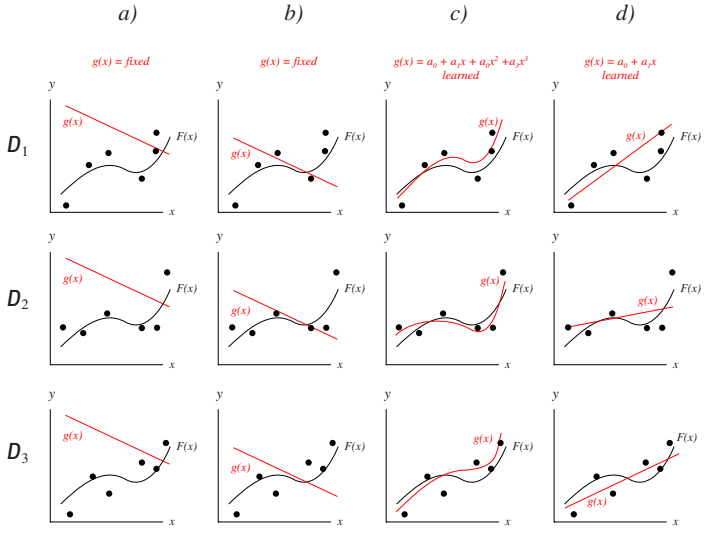
$$\mathop{\mathrm{E}}_{x,y,\mathcal{D}}\left[\left(\hat{f}(x;\,\mathcal{D})-y\right)^2\right]=$$

$$\underbrace{\mathop{\mathrm{E}}_{x,\mathcal{D}}\left[\left(\hat{f}(x;\,\mathcal{D})-\bar{f}(x)\right)^2\right]}_{\text{Variance}}+\underbrace{\mathop{\mathrm{E}}_{x,y}\left[\left(\bar{f}(x)-\bar{y}(x)\right)^2\right]}_{\text{Bias}^2}+\underbrace{\mathop{\mathrm{E}}_{x,y}\left[\left(\bar{y}(x)-y\right)^2\right]}_{\text{Noise}}$$

Darts Example



Illustration



Questions to answer for yourself / Discuss with Friends

Repetition: What are the main components into which a model's error can be decomposed? Describe each component in your own words.

Ponder: Consider a situation where a model consistently makes the same types of errors across different datasets. Which component of the expected test error –bias, variance, or noise– likely contributes most to this consistent error?

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- **Noise:** How much error is due to the data itself and not the model.

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The bias

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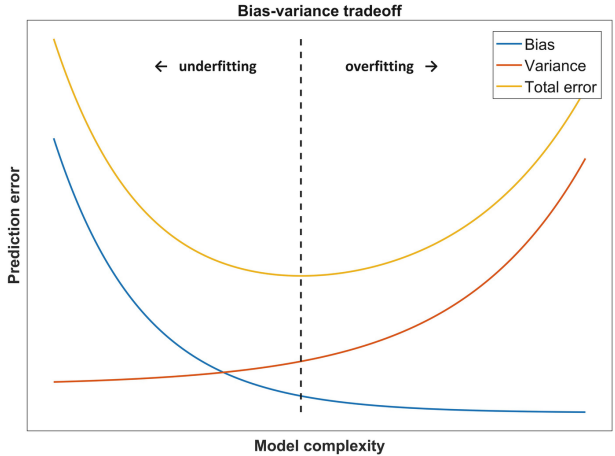
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Model complexity matters



Source: Dankers et al., 2018

Interpretation

- The choice of your model complexity will either:
  - increase the bias $\rightsquigarrow$ decrease the variance;
  - increase the variance $\rightsquigarrow$ decrease the bias.
- You cannot not control the noise.

Table 2: Understanding Variance, Bias, Noise

Term	High	Low
Variance	Risk of overfitting	Generalization
Bias	Risk of underfitting	Fitting
Noise	Challenging Task	Easy Task

Find a model complexity that has both a **Low Bias** (able to fit well) and a **Low Variance** (able to generalize).

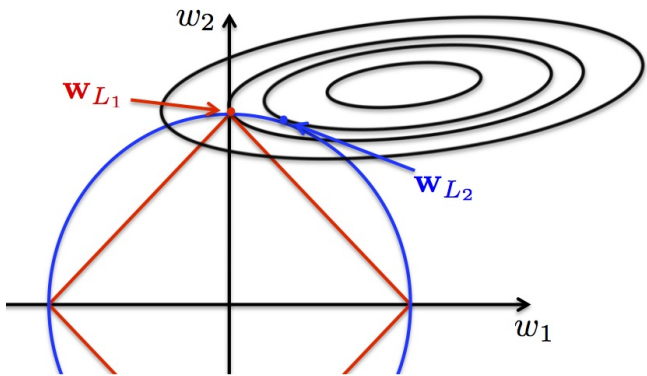
Weight Decay - L_1 / L_2 Regularization

- Add a penalty term to the empirical risk ($\alpha \in \mathbb{R}_+$):

$$\arg\min_{\theta} \left[\frac{1}{N} \sum_{i=1}^{N} \mathcal{L}\left(y_i,\, f(x_i;\, \theta)\right) \right] + \alpha\, \Omega(\theta)$$

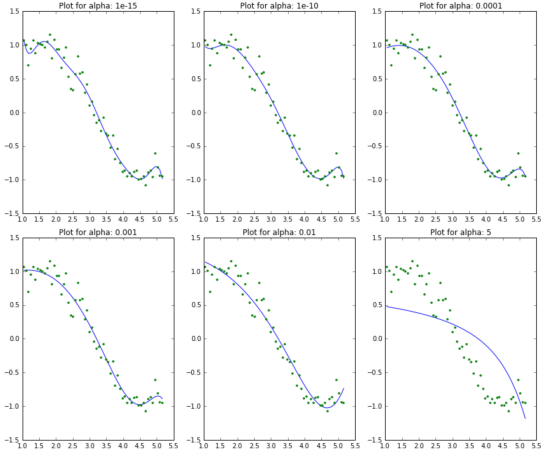
- The regularization penalizes high parameter values:
  - L_1: $\Omega(\theta) = \frac{1}{|\theta|} \sum_{k=1}^{|\theta|} |\theta_k|$
  - L_2: $\Omega(\theta) = \frac{1}{|\theta|} \sum_{k=1}^{|\theta|} \theta_k^2$

Illustration of the L_1 / L_2 Regularizations



Source: g2pi.tsc.uc3m.es

Regularizing a Polynomial Regression



$$f(x, \theta) = \sum_{j=0}^{15} \theta_j x^j$$ (Source: `www.analyticsvidhya.com`)

Questions to answer for yourself / Discuss with Friends

Repetition: In your own words, explain why adding a regularization term (like L_1 or L_2) helps prevent overfitting. What effect does regularization have on the model parameters?

Repetition: How would increasing the regularization strength (α) affect a model's performance? Which trade-offs should you consider when adjusting this parameter for L_1 / L_2 regularization?

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Regularization is a technique for ”restricting” a model’s complexity. Both L_1 and L_2 regularization try to make the learnable weights as small as possible.

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Regularization is always a tug-of-war with the main loss function. A higher weight for the regularization term necessarily means that the actual task will be less important.

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Take-Home Recipe

- The Bias-variance Tradeoff is a fundamental concept in ML
- Low bias and Low variance models are demanded
- Models should be regularized when exhibiting High Variance
  - Do not over-regularize to the point of having a very high bias
  - Remember the aim is accurate generalization